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PERFORMANCE SPECIFICATION

PUMP, SUBMERSIBLE, BILGE, ELECTRIC-MOTOR-DRIVEN

This specification is approved for use by the US Army Tank-automotive and Armaments Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a 24 volt electric-motor-driven, submersible bilge pump for moving water.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E/BLUE, Warren, MI 48397-5000, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

AMSC N/A FSC 4320

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

STANDARDS

DEPARTMENT OF DEFENSE

MIL-STD-202	- Test Methods for Electronic and Electrical Component Parts.
MIL-STD-461	- Control of Electromagnetic Interference Emissions and Susceptibility.
MIL-STD-462	- Electromagnetic Interference Characteristics, Measurement of.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation (see 6.2).

DRAWINGS

ARMY

12369000	- Chemical Agent Resistant Coatings (CARC) Paint Systems Index.
12292104	- Pump Unit, Submersible.

(Copies of these drawings are available from the U.S. Army Tank-automotive and Armaments Command, AMSTA-TR-E/BLUE, Warren, MI 48397-5000.)

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2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/ASQC Z1.4 - Sampling Procedures and Tables for Inspection by Attributes (DoD Adopted).

(Application for copies should be addressed to American National Standard Institute, 11 West 42nd Street, New York, NY 10036.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM B842 - Electrodeposited Coatings for Zinc Iron Alloy Deposits.

(Copies of ASTM publications may be obtained from the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)

SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)

AMS-2471D - Anodic Treatment of Aluminum Alloys, Sulfuric Acid Process, Undyed Coating (DoD Adopted).

(Application for copies should be addressed to Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.3.

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3.2 Materials. Materials used shall be in accordance with the manufacturer's materials specifications. The materials shall be capable of meeting all the operational and environmental requirements specified herein. Asbestos and cadmium materials shall not be used in any form in any part of the vehicle. No item, part or assembly shall contain radioactive materials in which the specific activity is greater than 0.002 microcurie per gram or activity per item equals or exceeds 0.01 microcuries. Recovered materials shall be used to the maximum extent practicable (see 4.8.1 and 4.8.2).

3.3 Design and construction. All pumps shall conform to the manufacturer's design and construction except for outline dimensions. Attaching hole dimensions and locations, and porting holes inlet/outlet dimensions and locations shall conform to drawing 12292104 (see 4.8.1 and 4.8.2).

3.3.1 Interchangeability. To provide for interchangeability, all assemblies and components having the same part number shall be functionally and dimensionally interchangeable (see 4.8.2).

3.3.2 Dissimilar metals. Except where necessary to complete an electrical circuit, contact between dissimilar metals, which would encourage galvanic action, shall be avoided (see 4.8.2).

3.3.3 Standard parts. Military standard parts shall be used wherever possible (see 4.8.2).

3.3.4 Electric motor. The pump shall have an enclosed waterproof electric motor which shall be rated at 28 volts direct current (Vdc) (see 4.8.1).

3.3.5 Steel components. Steel components shall be finished in accordance with ASTM B842 class I, Grade 3 (see 4.8.1).

3.3.6 Aluminum components. Aluminum components shall be anodized in accordance with SAE AMS 2471D (see 4.8.1).

3.3.7 Corrosion resisting steel (CRES) components. CRES components shall show no evidence of rust or staining after being tested in accordance with 4.7.1.

3.4 Performance.

3.4.1 Overall performance. The pump output (flow) and current consumption shall be as shown in figure 1 when operated at 28 ± 0.05 Vdc at any indicated head pressure (see 4.8.3.1).

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3.4.2 Water level. The pump shall start to deliver water when the depth of water is not more than 7.5 centimeters (cm) (3 inches), above the pump mounting surface. The pump may stop delivering water when the depth of water is not more than 7 cm (2.75 inches), above the pump mounting surface (see 4.8.3.2).

3.4.3 Pumping capacity. The pump shall deliver not less than 125 liters per minute (L/m) (33 gallon per minute (gal/min)) against a 2.3 meter (m) (7.5 foot) static pressure head through a 3.55 ± 0.15 cm (1.40 ± 0.06 inches) inside diameter outlet (see 4.8.3.3).

3.4.4 Dry-run capability. The pump shall retain not less than 90 percent (%) of the pumping capacity specified in 3.4.3, after 25 hours of dry running (see 4.8.3.4).

3.4.5 Air-lock limitation. The pump shall not air-lock for more than 20 seconds under any pumping condition (see 4.8.3.5).

3.4.6 Dielectric strength. The pump shall show no evidence of puncture of solid insulation, continuing sparkover, breakdown of the dielectric material, or damage to the insulation sufficient to cause failure of the motor; after being exposed to a 500 V root-mean-square (rms), 60 hertz (Hz) potential for 1 minute (see 4.8.3.6).

3.4.7 Starting capacity. The motor of the pump shall be capable of starting with 13.5 to 15 Vdc, applied at -1 degree Celsius (°C) (30 degrees Fahrenheit (°F)) (see 4.8.3.7).

3.4.8 Environmental.

3.4.8.1 High temperature. The pump shall meet the overall performance requirements specified in 3.4.1 at an ambient temperature of 65 ± 3°C (149 ± 5°F) after being subjected to storage at a temperature of 71°C (160°F) (see 4.8.3.8.1).

3.4.8.2 Low temperature. The pump shall meet overall performance requirements specified in 3.4.1 at ambient temperatures of 2 ± 2°C (36 ± 4°F) after being subjected to storage at a temperature of -54 ± 3°C (-65 ± 5°F) (see 4.8.3.8.2).

3.4.8.3 Shock. The pump shall show no evidence of deformed, cracked or broken parts and shall meet the requirements of 3.4.3 after being subjected to sawtooth waveform shock pulses of 40 gravity units (g) and 11 milliseconds (ms) time duration along each of the three major axes (see 4.8.3.8.3).

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3.4.8.4 Vibration. The pump shall show no evidence of deformed, cracked or broken parts and shall meet the requirements of 3.4.3 after exposure to vibration levels, in each of the three major axes, of 1.5 g and 4.2 g in a frequency range of 5 to 500 to 5 Hz (see figure 2 and 4.8.3.8.4).

3.4.8.5 Electromagnetic interference. The pump shall meet the electromagnetic interference requirements of MIL-STD-461. Limits RE05 and CE07 (see 4.8.3.8.5).

3.4.9 Endurance. The pump shall retain not less than 90% of pumping capacity specified in 3.4.3 after being operated for 100 hours (see 4.8.3.9).

3.5 Finish. Unless otherwise specified (see 6.2), exposed ferrous and aluminum surfaces shall be cleaned, pretreated, primed, and painted in accordance with an applicable CARC paint system selected from Drawing 12369000 for external application. Unless otherwise specified, color of paint final topcoat shall be green 383 (see 4.8.2).

3.5.1 Hydrogen embrittlement. Ferrous metal parts with a Rockwell hardness of C40 or greater shall be baked for hydrogen embrittlement relief for parts receiving pickle cleaning or electroplating operations within four hours of cleaning and/or electroplating operation (see 4.7.2).

3.6 Identification marking. Unless otherwise specified (see 6.2), a nameplate shall be attached to the pump housing in the manner specified on Drawing 12292104. The legend contained on the plate shall include the following (see 4.8.2):

a. Bilge pump
b. 28 Vdc
c. Manufacturer's identification
d. Serial number
e. Military part number
f. Federal stock number
g. Manufacturer's part number
h. Date of manufacture
i. Contract number
j. US.

3.7 Workmanship. Workmanship shall be of a quality which will assure a product free of cracked, broken, or rusted parts or components; scratched finish, peeled or blistered paint; or other defects which may affect appearance or performance (see 4.8.2).

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4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
   1. Preproduction inspection (see 4.3.1).
   2. Initial production inspection (see 4.3.2).
b. Conformance inspections (see 4.4).
   1. Examination (see 4.4.2).
   2. Test (see 4.4.3).
c. Control tests (see 4.5).

4.2 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed under the following standard (room) ambient conditions:

a. Air temperature: $25 \pm 6°C$ ($77 \pm 10°F$).
b. Relative humidity: (uncontrolled room ambient).
c. Atmospheric pressure: (site pressure).

4.2.1 Temperature stabilization. Test specimens shall be stabilized at room temperature for 1 hour before tests are begun.

4.2.2 Test set-up. Unless otherwise specified herein, the pump shall be positioned with the axes of rotation horizontal during testing to simulate in-use operation.

4.2.3 Environmental tests. When specified herein, the pump shall be operated at ambient conditions prior to proceeding with any of the environmental tests to obtain a baseline for determining satisfactory operation of the item during and after the tests, as applicable.

4.3 First article inspection. When first article is required (see 3.1), first article inspection shall be performed on first article samples (see 6.2). Unless otherwise specified (see 6.2), first article inspection shall include the inspections specified in table I.

4.3.1 Preproduction inspection. When specified (see 6.2), the preproduction sample shall consist of three pumps (specimens A, B, and C). Preproduction inspection shall consist of inspection as specified in table I. Performance tests shall be conducted in the order specified in table II.

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TABLE I. Classification of inspections.

Title	Requirements	Inspection	First article	Conformance			
				Examination	Test groups		Control
					A	B	
Materials and construction	3.2, 3.3, and 3.3.4 thru 3.3.7	4.8.1	X				X
Defects (see 4.4.2 and table III)	3.3 thru 3.3.3, 3.5, 3.6, and 3.7	4.8.2	X	X			
Overall performance	3.4.1	4.8.3.1	X				X
Water level	3.4.2	4.8.3.2	X				X
Pumping capacity	3.4.3	4.8.3.3	X			X	X
Dry-run capability	3.4.4	4.8.3.4	X				X
Air-lock limitation	3.4.5	4.8.3.5	X				X
Dielectric strength	3.4.6	4.8.3.6	X		X		
Starting capability	3.4.7	4.8.3.7	X		X		
High temperature	3.4.8.1	4.8.3.8.1	X				X
Low temperature	3.4.8.2	4.8.3.8.2	X				X
Shock	3.4.8.3	4.8.3.8.3	X				X
Vibration	3.4.8.4	4.8.3.8.4	X				X
Electromagnetic interference	3.4.8.5	4.8.3.8.5	X				X
Endurance	3.4.9	4.8.3.9	X				X

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TABLE II. Order of tests.

Specimen	Title	Test
A	Pumping capacity	4.8.3.3
	Endurance	4.8.3.9
	Air-lock limitation	4.8.3.5
	Shock	4.8.3.8.3
	Vibration	4.8.3.8.4
B	Overall performance	4.8.3.1
	Water level	4.8.3.2
	Pumping capacity	4.8.3.3
	Air-lock limitation	4.8.3.5
	Dielectric strength	4.8.3.6
C	Pumping capacity	4.8.3.3
	Dry-run capability	4.8.3.4
	High temperature	4.8.3.8.1
	Low temperature	4.8.3.8.2
	Air-lock limitation	4.8.3.5
	Shock	4.8.3.8.3
	Vibration	4.8.3.8.4
	Starting capability	4.8.3.7
	Electromagnetic interference	4.8.3.8.5

4.3.2 Initial production inspection. The Government shall select three pumps from the first lot produced under the production contract for initial inspection. Initial production units shall be inspected as specified in tables I and II.

4.3.3 First article inspection failure. Deficiencies found during, or as a result of, first article inspection shall be cause for rejection of the first article sample until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiency. Any deficiency found during, or as a result of, first article inspection shall be evidence that all items already prior to completion of the first article inspection are similarly deficient unless contrary evidence satisfactory to the contracting officer is furnished by the contractor. Such deficiencies on all items shall be corrected by the contractor. The Government will not accept products until first article inspection is completed to the satisfaction of the Government.

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4.4 Quality conformance inspections.

4.4.1 Sampling.

4.4.1.1 Lot formation. An inspection lot shall consist of all the pumps from an identifiable production period, from one manufacturer, submitted at one time for acceptance.

4.4.1.2 Sampling for examination. Samples for quality conformance examination shall be selected in accordance with ANSI/ASQC Z1.4.

4.4.1.3 Sampling for group A tests. Samples of group A tests shall be selected in accordance with ANSI/ASQC Z1.4.

4.4.2 Examination.

4.4.2.1 Classification of defects. Each sample selected in accordance with 4.4.1.2 shall be examined and shall be free from any defects as listed in table III.

TABLE III. Classification of defects.

Category	Defect	Method of examination
Critical	None	
Major		
101	Use of cadmium, asbestos or materials with dangerous radioactivity (see 3.2).	Visual and SIE 1/
102	Nonconformance in design and construction (see 3.3, 3.3.2).	SIE
103	Dimensions affecting interchangeability out of tolerance (see 3.3 and 3.3.1).	Visual and SIE
104	Faulty workmanship affecting performance (see 3.7).	SIE
Minor		
201	Dimensions not affecting interchangeability out of tolerance (see 3.3).	Visual and functional
202	Improper finish (see 3.5).	Visual
203	Improper marking (see 3.6).	Visual
204	Faulty workmanship affecting appearance (see 3.7).	Visual

1/ SIE = Standard Inspection Equipment

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4.4.3 Tests.

4.4.3.1 Group A. Samples selected in accordance with 4.4.1.3 shall be subjected to the quality conformance tests specified in table I and shall be free from defects..

4.4.3.2 Group B. Each pump shall be subjected to the quality conformance test specified in table I and shall be free from defects.

4.4.3.2.1 Failure. Failure of any pump to pass any of the specified quality conformance tests shall be cause for the Government to refuse acceptance to the production quantity represented, until action taken by the contractor to correct defects and prevent recurrence has been approved by the Government.

4.5 Control tests. When specified (see 6.2), control tests shall be conducted on 2 pumps from each lot of 100 units consecutively produced, except that not more than 4 pumps shall be tested in a 30-day period. The pump shall be subjected to the control tests specified in table I.

4.6 Failure. Failure of any pump to pass any of the specified control tests shall be cause for the Government to refuse acceptance of the production quantity represented, until action taken by the contractor to correct defects and prevent recurrence has been approved by the Government.

4.7 Method of testing.

4.7.1 Corrosion resisting steel components (CRES). To determine conformance to 3.3.7, actual pump components shall be subjected to the salt spray test specified by ASTM B117 for a minimum of 2 hours using a 5% salt solution. Prior to this test, the samples shall be immersed in a solution of nitric acid and sodium dichromate to dissolve surface contaminants resulting from fabrication processes such as machining, etc. Samples shall be selected in accordance with ANSI/ASQC Z1.4.

4.7.2 Hydrogen embrittlement. To determine conformance to 3.5.1, parts shall be heat treated after phosphating for 8 hours at 99 - 107°C (210 - 225°F) or must be held for 240 hours at room temperature to relieve any embrittlement due to hydrogen. Parts having a hardness of RC48 or higher shall not be subjected to phosphating. The test for the affectiveness of the procedure used to control embrittlement lies in the subsequent service use of the material. Applicable control test(s) for hydrogen embrittlement can be selected by analogy to the particular service in which the processed article shall be used. Samples shall be selected in accordance with ANSI/ASQC Z1.4.

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4.8 Methods of inspection.

4.8.1 Materials and construction. Conformance to 3.2, 3.3, and 3.3.4 through 3.3.6 shall be determined by inspection of contractor records providing proof or certification that design, construction, processing, and materials conform to requirements. Applicable records shall include drawings, specifications, design data, receiving inspection records, processing and quality control standards, vendor catalogs and certifications, industry standards, test reports, and rating data.

4.8.2 Defects. Conformance to 3.2, 3.3, 3.3.1 through 3.3.3, 3.5, 3.6 and 3.7 shall be determined by examination for the defects listed in table III. Examination shall be visual, functional, or by measurement with standard inspection equipment.

4.8.3 Performance tests.

4.8.3.1 Overall performance. To determine conformance to 3.4.1, the pump shall be operated against a static pressure head up to 6.4 m (20.9 feet) at 28 ± 0.05 Vdc. The output volume and current consumption shall be measured and recorded at various head pressures.

4.8.3.2 Water level. To determine conformance to 3.4.2, the pump shall be operated submerged and the water level allowed to drop until the water discharge stops. With the pump running, raise the water level and measure and record the water level relative to the pump mounting surface at which the pump begins to discharge water. Allow the water level to drop and measure and record the level at which the water discharge stops.

4.8.3.3 Pumping capacity. To determine conformance to 3.4.3, the pump shall be operated and the output shall be measured, under the following test conditions:

a. The static pressure head shall be 2.3 m (7.5 feet).
b. The voltage at motor terminals shall be 28 ± 0.05 Vdc.
c. The ambient air temperature shall be 25 ± 6°C (77 ± 11°F).
d. The specific gravity of water being pumped shall be no greater than 1.03.
e. The water temperature shall be 25 ± 6°C (77 ± 11°F).

4.8.3.4 Dry-run capability. To determine conformance to 3.4.4, the pump shall be dry-run for 25 hours under the following conditions:

a. The voltage at motor terminals shall be 26 ± 0.05 Vdc.
b. The ambient air temperature shall be 65 ± 6°C (149 ± 10.8°F).

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c. The pump shall be attached to a discharge line as defined in figure 3.
d. There shall be no maintenance or adjustments to the pump permitted during the test.

After the test, the pump capacity shall be determined in accordance with 4.8.3.3. The pumping capacity shall be not less than 90% of that specified in 3.4.3.

4.8.3.5 Air-lock limitation. To determine conformance to 3.4.5, occurrence of air-lock, if any, shall be timed and recorded during all tests and shall be not more than as specified under any pumping condition.

4.8.3.6 Dielectric strength. To determine conformance to 3.4.6, the pump shall be tested in accordance with the method 301 of MIL-STD-202, where the test voltage shall be 500 V rms at 60 Hz and applied for 1 minute between ground and each winding or electrical circuit element. The voltage source shall be capable of delivering 1 milliampere (mA) to a 500 ohm resistive load. Brushes shall be raised from commutator and winding ungrounded. The filter capacitor in the electric connector shall be disconnected during this test. After the test, the pump shall be checked for any damage or malfunction.

4.8.3.7 Starting capability. To determine conformance to 3.4.7, apply a voltage in the range specified to the pump electrical terminals, at the specified temperature, and observe and record whether the pump starts.

4.8.3.8 Environmental.

4.8.3.8.1 High temperature. To determine conformance to 3.4.8.1, the pump shall be tested as specified in 4.8.3.3 except that the pump and the water shall be stabilized at 65 ± 3°C (149 ± 11°F).

4.8.3.8.2 Low temperature. To determine conformance to 3.4.8.2, the pump shall be stabilized at -54 ± 3°C (-65 ± 5°F) for 6 hours. The pump and water shall then be stabilized at 2 ± 2°C (36 ± 4°F) and tested as specified in 4.8.3.3.

4.8.3.8.3 Shock. To determine conformance to 3.4.8.3, the pump shall be mounted in a position to simulate its actual use and three shocks in each direction shall be applied along its three mutually perpendicular axis (total of 18 shocks). The shock pulse shape shall be a sawtooth waveform in accordance with figure 4 of an amplitude and a time duration as specified. After the test, the pump shall be tested as specified in 4.8.3.3 (see 4.2.3).

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4.8.3.8.4 Vibration. To determine conformance to 3.4.8.4, the pump shall be mounted, in a manner that will simulate its actual use, and vibrated along each axis in accordance with the vibration levels specified in figure 2. The sweep time shall be 15 minutes for the sweep frequency range of 5 to 500 to 5 Hz. The cycling time shall not exceed 3 hours in each axis. The frequency of applied vibration shall be swept over the specified range logarithmically in accordance with figure 5. The specified sweep time is that of an ascending plus a descending sweep and is twice the ascending sweep time shown on figure 5 for the specified range. When the pump resonances below 5 Hz are measured, the test curve shall be extended to 2 Hz and the sweep time shall be 18 minutes (2 to 500 to 2 Hz). There shall be no operational tests required during vibration. After the test, the pump shall be tested as specified in 4.8.3.3 (see 4.2.3).

4.8.3.8.5 Electromagnetic interference. To determine conformance to 3.4.8.5, the pump shall be tested for electromagnetic interference limits RE102 and CE102 as defined in MIL-STD-461. These limits shall be tested in accordance with MIL-STD-462.

4.8.3.9 Endurance. To determine conformance to 3.4.9, the pump shall pump water containing sodium chloride, sand and clay mixed in the following proportions and under the following conditions:

a. Mix 95 parts (by volume) fresh water and 5 parts sodium chloride.
b. For each 189 liters (L) (50 gallon) of solution in a, add 2.3 kg (5 lbs) of clay (Bentonite) and 0.9 kg (2 lbs) of sand. The sand shall be 0.01 to 1.0 mm (0.0004 - 0.04 inches) in diameter with the majority of grains 0.15 to 0.30 mm (0.006 - 0.012 inches) in diameter.
c. Circulation and agitation shall be such that the sand and clay remain in suspension throughout the test.
d. The test conditions shall be as specified in 4.8.3.3.

4.8.3.9.1 Procedure. The endurance test shall be performed as follows:

a. Water, as specified in 4.8.3.9, shall be pumped for 6 hours at 2.3 m (7.5 feet) water head and a flow of not less than 125 L/m (33 gal/min).
b. The pump shall rest for not less than 30 minutes.
c. Three cycles shall be performed as follows:
    1. Three hours of pumping, as specified in step a.
    2. Twenty minutes of dry-run.
    3. A rest of not less than 30 minutes.
    4. Three hours of pumping, as specified in step a.
    5. A rest of not less than 30 minutes.

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d. The above steps (a through c) shall be performed four times. After the test, the pump capacity shall be determined in accordance with 4.8.3.3. The pumping capacity shall not be less than 90% of that specified in 3.4.3.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature which may be helpful, but is not mandatory.)

6.1 Intended use. The pump covered by this specification is intended to expel water and contaminants (sea water, fuels, and lubrications) from the hulls of combat vehicles.

6.2 Acquisition requirements. Acquisition documents must supply the following:

a. Title, number, and date of the specification.
b. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1, 2.2.2, and 2.3).
c. If first article inspection is required (see 3.1).
d. If finish should be other than specified (see 3.5).
e. If identification marking should be other than as specified (see 3.6).
f. If inspection conditions should be other than as specified (see 4.2).
g. Type of first article sample, the number of samples to be inspected, and (when applicable) the specific tests for each sample (see 4.3).
h. First article inspection, if other than as specified (see 4.3).
i. Preproduction sample specified (see 4.3.1).
j. If control test is required and if control test sample size should be other than as specified (see 4.5).
k. Packaging requirements (see 5.1).

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6.3 Supersession data. This specification supersedes MIL-P-62415(AT), dated 22 June 1989.

6.4 Subject term (key word) listing.

Draw fluid
Horizontal flow
Water
Waterproof

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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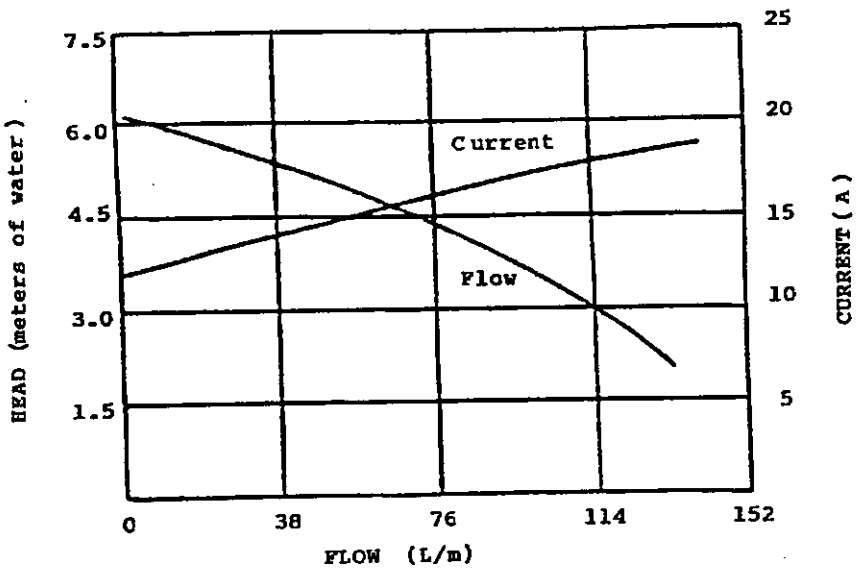
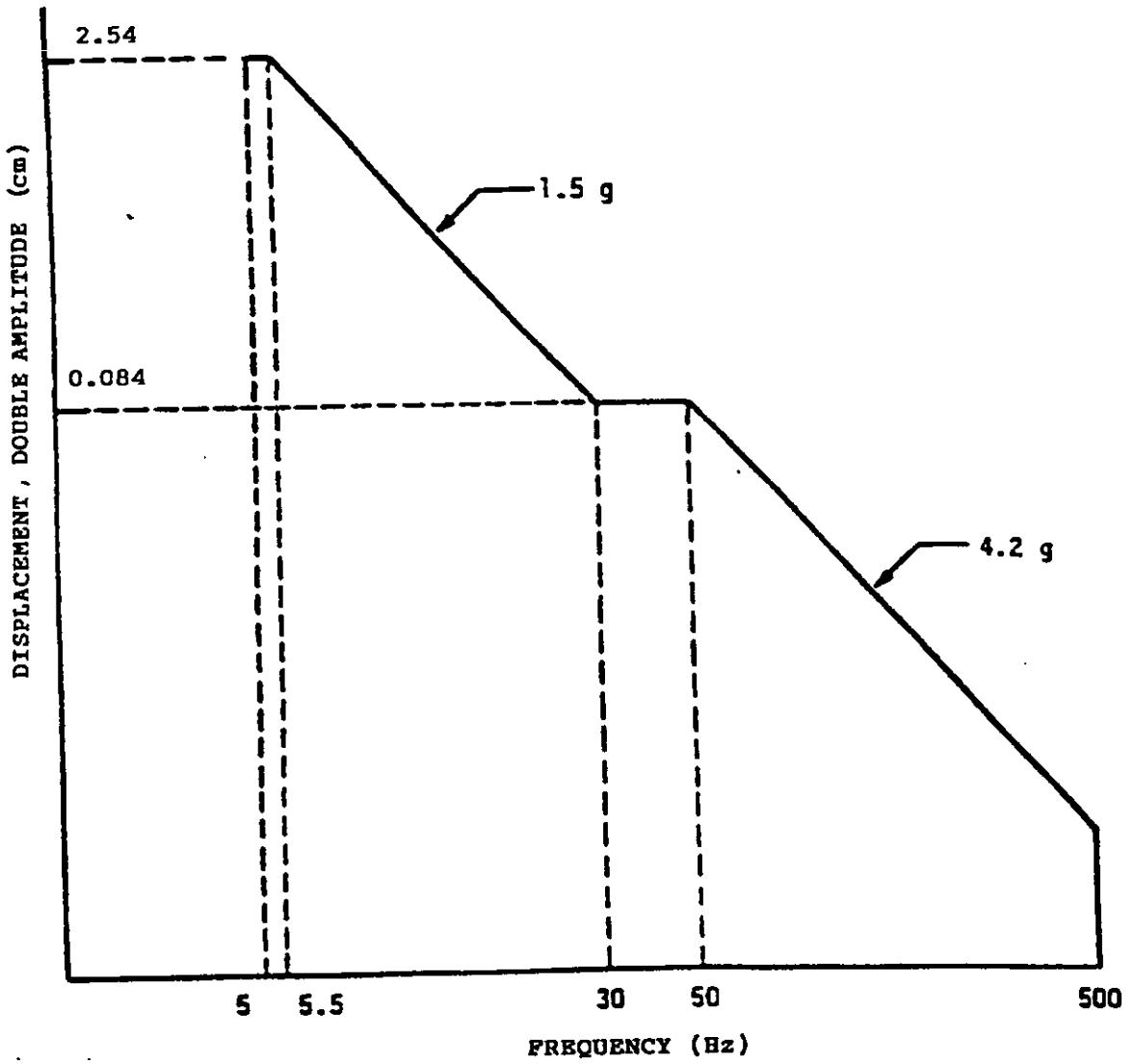


FIGURE 1. Performance data.

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NOTES:

1. Acceleration levels: ± 9 g.
2. The curve shall be extended to 2 Hz when test item resonances below 5 Hz are expected.

FIGURE 2. Vibration curve.

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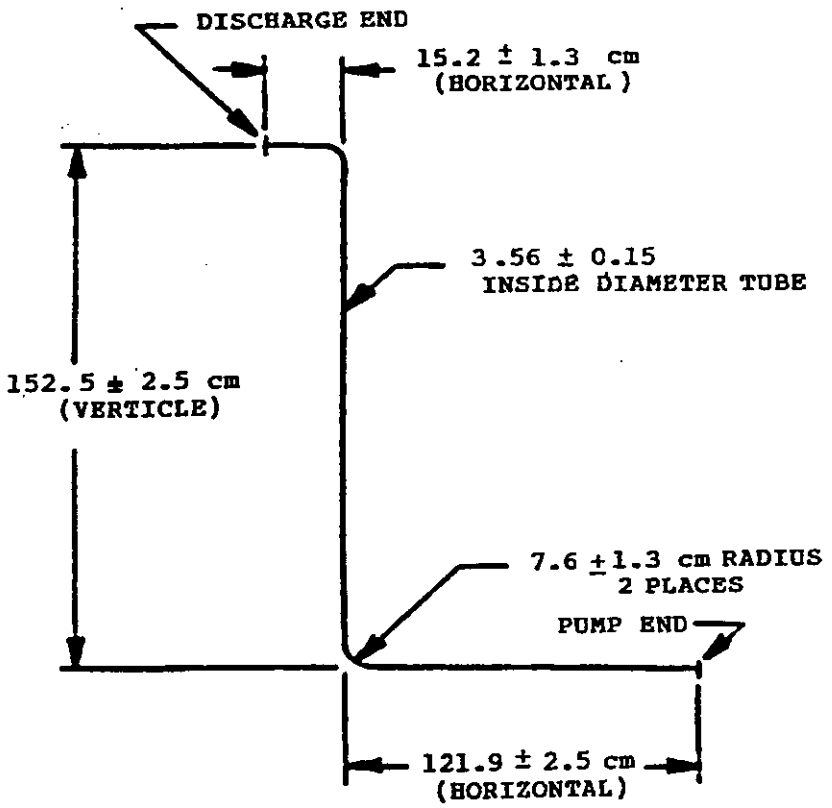
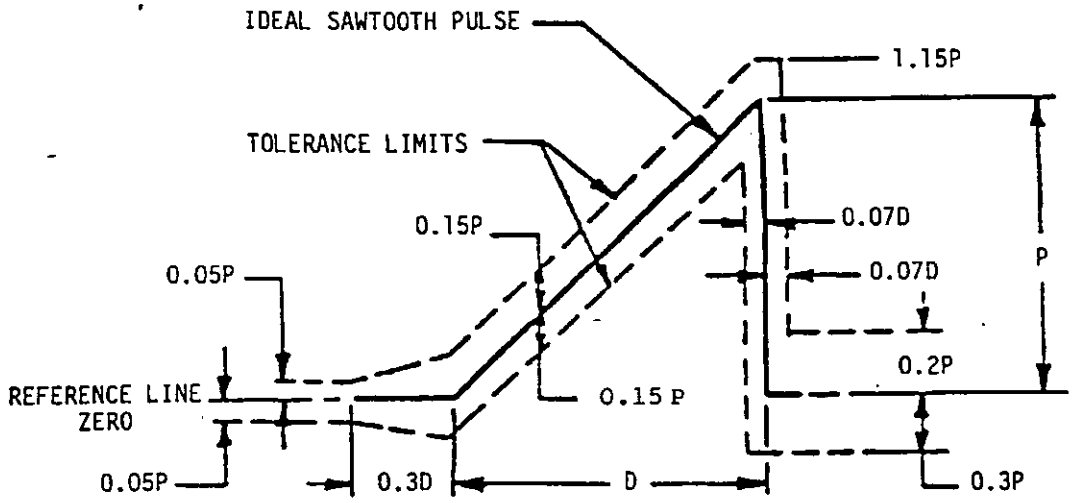


FIGURE 3. Discharge tube for dry-run test.

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P = Peak value = 40 g
D = Nominal duration = 11 ms

NOTE: The oscillogram shall include a time about 3D long with a pulse located approximately in the center. The peak acceleration magnitude of the sawtooth pulse is P and its duration is D. The measured acceleration pulse shall be contained between the broken line boundaries and the measured velocity change (which may be obtained by integration of the acceleration pulse) shall be within the limits of $V_i \pm 0.1\ V_i$ where V_i is the velocity-change associated with the ideal pulse which equals 0.5 PD. The integration to determine velocity change shall extend from 0.4D before the pulse to 0.1D after the pulse.

FIGURE 4. Terminal-peak sawtooth shock pulse configuration and tolerance limits.

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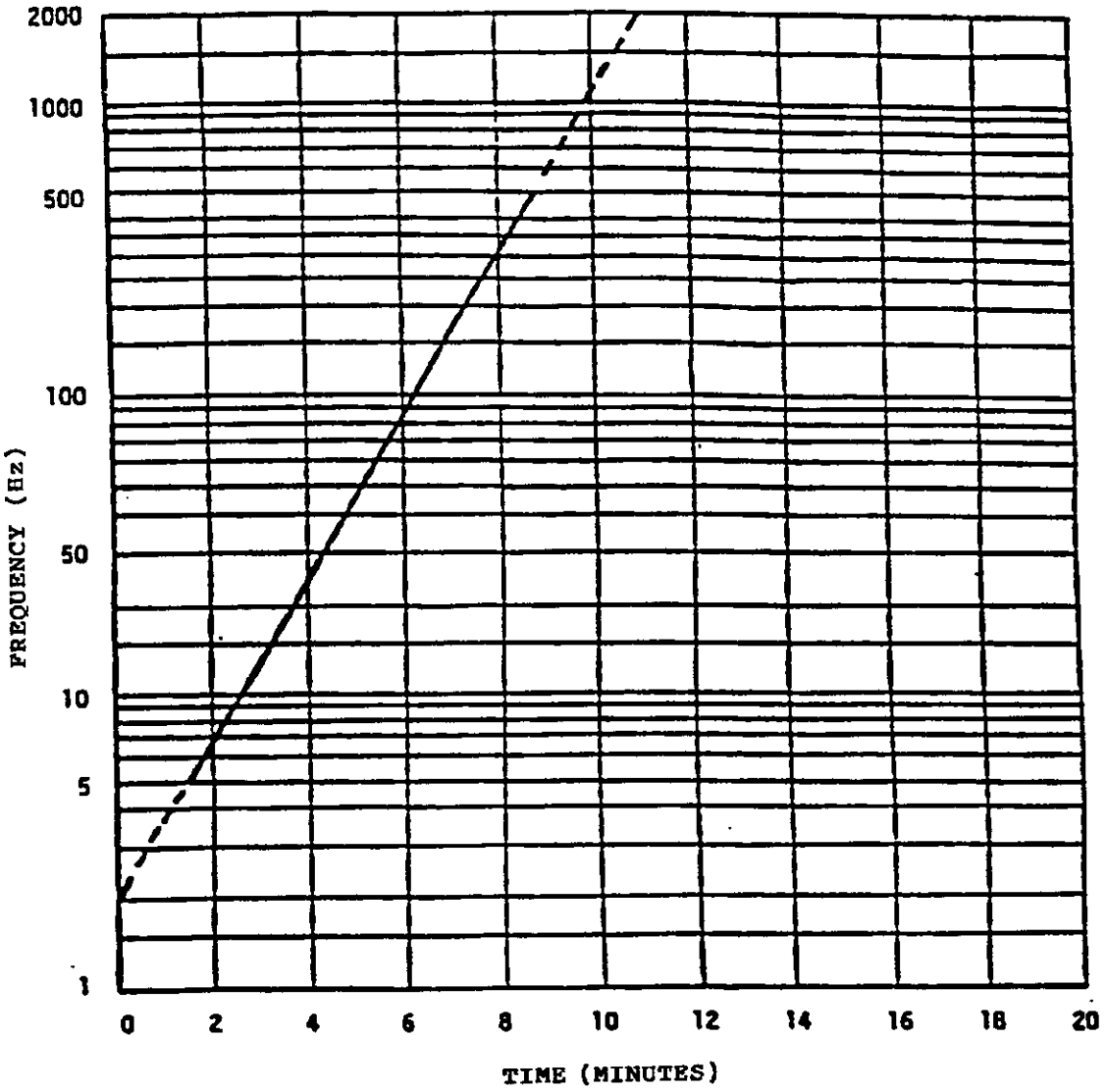


FIGURE 5. Logarithmic sweep.

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	MIL-PRF-62415A(AT)	960311

3. DOCUMENT TITLE

Pump, Submersible, Bilge, Electric-Motor-Driven

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

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